- Fungi and Bacteria are decomposers. They breakdown organic matter into simple inorganic substances.
- Virus represents dormant life. They are metaboli- cally inactive if they are outside a host body. They are not decomposers. They invade host cells and use their nucleus (DNA machinery) to carry out their life processes.

15. Based on the replacement period, a nutrient cycle is referred to as Perfect or Imperfect cycle.
    - A perfect nutrient cycle is one in which nutrients are replaced as fast as they are utilised.
    - Most gaseous cycles are generally considered as perfect cycles. In contrast sedimentary cycles are considered relatively imperfect, as some nutrients are lost from the cycle and get locked into sediments and so become unavailable for immediate cycling.
    - Based on the nature of the reservoir, a nutrient cycle is referred to as Gaseous or Sedimentary cycle
    - Gaseous Cycle: the reservoir is the atmosphere or the hydrosphere—water cycle, carbon cycle, nitrogen cycle, etc., and
    - Sedimentary Cycle: the reservoir is the earth's crust (soluble elements mostly found in earth's crust)—phosphorous cycle, sulphur cycle, calcium cycle, magnesium cycle, etc.

16.
    - The amount of nitrogen fixed by man through the industrial process has far exceeded the amount fixed by the Natural Cycle.
    - As a result, nitrogen fixed by man has become a pollutant which can disrupt the balance of nitrogen. It may lead to Acid rain, Eutrophication and Harmful Algal Blooms.
    - Oxides of nitrogen fall on earth as acid rain. Acidic rain is a complex mixture of nitrous, nitric, sulphurous and sulfuric acids which all combine to lower the pH.
    - Amount of excessive fixed nitrogen being added to soil in form of fertilizers adds nitrogen to the soil and pollutes water bodies near through run of and leaching.